MIL-P-46036B(MR)
23 February 1973
SUPERSEDING
MIL-P-46036A(MR)
18 October 1966

MILITARY SPECIFICATION

PLASTIC SHEETS, RODS, TUBES AND DISCS,

POLYCHLOROTRIFLUOROETHYLENE

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic sheets, rods, tubes and discs prepared from polychlorotrifluoroethylene.

1.2 Classification. Polychlorotrifluoroethylene sheets, rods, tubes and discs shall be of the following types, as specified (see 6.2).

Type I - General
Type II - Electrical

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Fiberboard
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, Waterproof

FSC 9330, 9390, 4710, 4720

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3. REQUIREMENTS

3.1 Material. The material furnished under this specification shall consist of polychlorotrifluoroethylene homopolymer or copolymers containing not less than 95 percent polychlorotrifluoroethylene. Material reprocessed from fabricated articles or waste shall not be included.

3.2 Property values. The plastic sheets, rods, tubes, and discs shall conform to the property values specified in tables I, II and III, as applicable, when tested as specified in the applicable procedure of 4.3.

Table I. Property values for all shapes and types

Property	Value	Paragraph reference to applicable test
Specific gravity, 23/23°C (73.4/73.4°F), minimum	2.10	4.3.2
Tensile strength, pounds per square inch (psi), minimum	4500	4.3.3
Deformation under load, (at 250 lb, 24 hr, 70 ± 1°C), percent, maximum	10	4.3.4

Table II. Additional property values for type II

Property	Value	Paragraph reference to applicable test
Dielectric constant, maximum		
at 1 KHZ	2.70	
at 1 MHZ	2.50	4.3.5
Dissipation factor, maximum		
at 1 KHZ	0.030	
at 1 MHZ	0.012	4.3.5

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Table V - Diameter tolerance for molded rods

Nominal diameter, inches	Tolerance, inches, plus and minus
0.500 to 1.000 inclusive (incl.)	0.020
Over 1.000 to 2.000 incl.	0.025
Over 2.000 to 3.500	0.035

Table VI - Diameter tolerances for molded tubes[1/]

Nominal outside diameter, inches	Tolerance, inches					
	Inside dia.		Outside dia.		Wall thickness	
	Plus	Minus	Plus	Minus	Plus	Minus
1.000 to 2.000 inclusive (incl).	0.000	0.060	0.060	0.000	0.125	0.000
Over 2.000 to 5.000 incl.	0.000	0.188	0.125	0.000	0.188	0.000
Over 5.000 to 8.000 incl.	0.000	0.188	0.188	0.000	0.250	0.000

1/ The wall thickness and either the outside diameter or the inside diameter, as specified, shall meet the tolerances.

Table VII - Diameter tolerance for extruded rods (unground)

Nominal diameter, inches	Tolerance, inches, plus and minus
0.060 and under	0.005
Over 0.060 to 0.125 inclusive (incl.)	0.008
Over 0.125 to 0.250 incl.	0.010
Over 0.250 to 0.500 incl.	0.015
Over 0.500 to 1.000 incl.	0.020
Over 1.000 to 2.000 incl.	0.025

Table VIII - Inside diameter tolerance for extruded tubes

Nominal inside diameter, inches	Tolerance, inches, plus and minus
0.060 and under	0.004
Over 0.060 to 0.125 inclusive (incl.)	0.006
Over 0.125 to 0.250 incl.	0.008
Over 0.250 to 0.500 incl.	0.010
Over 0.500 to 1.000 incl.	0.015
Over 1.000 to 2.000 incl.	0.020

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4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all sheets, rods, tubes and discs of the same type and thickness or diameter, as applicable, submitted for inspection at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced, subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification will govern. A suppliers certificate of compliance with 3.1 shall be furnished.

4.2.2 Examination of sheets, rods, tubes and discs.

4.2.2.1 Examination of sheets, rods, tubes and discs. Examination shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of packages of plastic sheets, rods, tubes and discs, as applicable, for examination in 4.2.2.1.1, 4.2.2.1.2, 4.2.2.1.3, 4.2.2.1.4 and in units of shipping containers for examination under 4.2.2.1.5.

4.2.2.1.1 Examination of sheets, rods, tubes and discs for defects in appearance, construction, and workmanship. The sample unit for this examination, specified in table XI, shall be one sheet, rod, tube or disc, as applicable. Not more than five sample units shall be taken from any one package.

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Table XII. Examination for dimensional defects Cont'd

Examine	Defect
Nominal inside diameter and tolerance for extruded tubes	Improper nominal inside diameter. Tolerances not in accordance with table VIII
Nominal wall thickness and tolerance for extruded tubes	Improper nominal wall thickness. Tolerances not in accordance with table IX.

4.2.2.1.3 Examination for defects in assembly. The sample unit for this examination, specified in table XIII, shall be one package of sheets, rods, tubes, or discs, as applicable.

Table XIII. Examination for defects in assembly

Examine	Defect
Assembly of sheets, rods, tubes, and discs	Not evenly stacked Sheets not interleaved to prevent adherence

4.2.2.1.4 Examination for defects in the count per package of sheets. The sample unit for this examination specified in table XIV shall be one package of sheets, rods, tubes, or discs, as applicable.

Table XIV. Examination for defects in the count per package

Examine	Defect
Sheets, rods, tubes and discs	Average count per package less than specified

4.2.2.1.5 Examination of preparation for delivery requirements. An examination shall be made in accordance with table XV to determine that packaging, packing and markings comply with the requirements of Section 5. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Examination Paragraph	Inspection Level	AQL
4.2.2.1.1	I	1.5
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-3	2.5
4.2.2.1.4	S-2	---
4.2.2.1.5	S-2	4.0

4.2.3 Testing. The sheets, rods, tubes, and discs shall be tested for the applicable characteristics listed in table I, in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 10 pounds of sheets, rods, tubes or discs, as applicable. The sample unit shall consist of sufficient sheet, rod, tube or disc, as applicable, to prepare all specimens required for testing. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens.

4.3 Test Methods.

4.3.1 Conditioning specimens and test conditions. Test specimens shall be conditioned in accordance with procedure A of ASTM D 618-61(1971), and shall be tested at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity.

4.3.2 Specific gravity. Three specimens shall be tested in accordance with ASTM D 792-66(1970). Method A1, A2, or A3 of ASTM D 792-66(1970) shall be used, and specimen preparation shall be in accordance with D 792-66(1970).

4.3.3 Tensile strength and elongation. The ultimate tensile strength and elongation shall be determined by testing five specimens in accordance with ASTM D 638-68, using the microtensile specimen of ASTM D 1708-66(1970). The initial jaw separation shall be 0.875 ± 0.005 inches and the speed of testing shall be 2.0 inches per minute. Shapes greater than 0.062 inches and up to 0.625 inches thick shall be machined to 0.062 ± 0.010 inches thick before cutting specimens. Shapes 0.625 inches thick and over shall have a slice somewhat thicker than 0.062 inches cut in a plane parallel to, and not less than 0.500 inches from the plane of one end of the shape. The slice shall be machined on both faces to 0.062 ± 0.010 inches thick, and the specimens cut from the machine slice. Tool marks shall be removed by light sanding in a longitudinal direction for all specimens reduced to specified thickness by machining.

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5.2.2 Level B. Packages of shapes shall be packed in domestic class or type shipping containers conforming to PPP-B-585 (class 1), PPP-B-601, or PPP-B-621. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of wood or wood-cleated containers shall not exceed 200 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. The shapes, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The plastic shapes are intended to be used primarily for making components having excellent chemical inertness, good thermal stability, and good electrical properties. The plastic is outstanding in resistance to water vapor transmission and as a gas and liquid barrier. The low temperature flexibility and inertness to oxidation permits applications for cryogenic equipment. Other applications include corrosion resistant linings and electrical and electronic components.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2 and 6.3).
(c) Shape and dimensions required (see 3.3 and 6.4).
(d) Centerless grinding of extruded rods, if required (see 3.3).
(e) Selection of applicable levels of packaging, and packing (see 5.1 and 5.2).
(f) Quantities for unit or intermediate packaging.
(g) Use of V3c or V3s fiberboard box, if required (see 5.2.2.1).

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Table XVII. Standard dimensions of extruded tubes and 72 inches long

Outside diameter, inches	Wall thickness, inches
1/8	1/32
1/4 to 1/2	1/32 to 1/16
1/2 to 2	1/16 and 1/8

Rods 1/16 through 1 inch inclusive in diameter may have 24 inch lengths.

Custodian:
Army - MR

Review interest:
Army - MU, EL, MI, GL

User interest:
Army - ME

Preparing Activity:
Army - MR

(Project No. 9330-A379)

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form, will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-P-46036B(MR), Plastic Sheets, Rods, Tubes and Discs, Polychlorotrifluoroethylene

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 83 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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